Particle Physics 2 Standard Model

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Supplementary track, 2010 Winter
Transcript-derived notes curated by LazyingArt LLC

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC.

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Chapter 1

Lecture 1: New Revolutions in Particle Physics: Standard Model

These notes follow Leonard Susskind's first lecture in *Particle Physics 2: Standard Model*, with curation by LazyingArt LLC. The lecture begins by delaying, once again, the promised particle list. That delay is not a digression. We already have two sides of the modern picture, particles and fields; before the catalogue can begin in earnest, we need the third side as well, namely force. The lecture therefore unfolds in a very definite order: first the field-language account of force, then the exchange-language account, and only then the first entries of the particle table.

1.1 Particles, Fields, and the Missing Third Leg

By this point in the course we have already built a good deal of machinery. We have fields and their quanta, wave equations and action principles, and the broad kinematic properties of particles: energy, momentum, spin, and the boson-fermion distinction. Fermions resist occupancy of the same state; bosons do not, and may pile up coherently into one state. We also know that products of fields in a Lagrangian encode processes, or vertices, in which particles come in and particles go out.

But something is still missing. Before we start naming the particles one by one, or rather group by group, the lecture insists that we complete a triangle of ideas:

- particles,
- fields for which those particles are quanta,
- forces associated with exchange processes involving those same quanta.

There is also a warning here, and it is worth preserving. The distinction between elementary and composite particles is not settled at the outset. We postpone it. For now we imagine that there are fundamental fields and that their quanta are elementary particles. The missing issue is different: if every field corresponds to a particle, and if every familiar field gives rise to a force, then there must also be a way to think about force directly in particle language.

That is the theme of the lecture. We are not going to be satisfied with Coulomb's law as a memorized formula. We want to see where a force comes from in the field description, and then see how the same phenomenon is retold when the field is regarded as a collection of quanta.

1.2 Force from Field Energy in Classical Electrodynamics

Susskind first closes the triangle in the classical language. Suppose we place two charges in space and ask for the force between them. One answer is to quote the inverse-square law. The lecture chooses a better answer: ask what the charges do to space.

They create electric fields. If the first charge produces $\mathbf{E}_1$ and the second produces $\mathbf{E}_2$, then the total field is

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2. \tag{1.1}$$

The transcript here is garbled in the details of a temporary charge-rescaled notation, so we keep only the stable mathematical core. The field energy is obtained schematically by squaring the electric field and integrating over space:

$$U_{\text{field}} \propto \int d^3x\, \mathbf{E}^2. \tag{1.2}$$

Substituting the sum of the two fields gives

$$U_{\text{field}} \propto \int d^3x\, (\mathbf{E}_1 + \mathbf{E}_2)^2 \tag{1.3}$$

$$\propto \int d^3x\, (\mathbf{E}_1^2 + \mathbf{E}_2^2 + 2\,\mathbf{E}_1 \cdot \mathbf{E}_2). \tag{1.4}$$

Now the lecture pauses and reads this expansion term by term. The first term is the self-energy of particle 1. The second is the self-energy of particle 2. Each is already part of the energy, hence the mass, of the corresponding isolated particle. Each is position-independent. For the present purpose, we set them aside.

The cross-term is different. It is not the energy of either particle separately. It exists only because both are present. If the charges are extremely far apart, the two Coulomb fields scarcely overlap and the integral is negligible. If they are brought closer, the overlap grows, and so does the new contribution. Thus the natural interaction energy is

$$U_{\text{int}}(r) \propto \int d^3x\, \mathbf{E}_1 \cdot \mathbf{E}_2. \tag{1.5}$$

The force is then read from the r-dependence of this interaction energy:

$$F(r) = -\frac{dU_{\text{int}}}{dr}. \tag{1.6}$$

If one performs the full calculation, one recovers the usual Coulomb behavior,

$$F_{\text{Coulomb}}(r) \propto \frac{q_1 q_2}{r^2}. \tag{1.7}$$

This is the lecture's first important payoff. The force law is now understood as coming from the portion of the field energy that depends on the relative arrangement of the two charges. One could say it this way: bring the charges together, the field in between them is distorted, and that distortion carries energy. The force is the derivative of that energy.

But if fields are nothing but collections of quanta, then this cannot be the end of the story. There must also be a particle-language account of the same force.

1.3 Molecular Exchange: Two Protons and One Electron

Before returning to electrodynamics in that second language, the lecture takes a detour to a system where the exchange picture is easier to see. Consider two protons and one electron. This is not chosen because it is the same as electrodynamics. It is chosen because the quantum mechanics is more intuitive.

If one proton is present and the other is taken very far away, the electron occupies the hydrogenic ground state centered on the first proton. The board first records this simply as a localized hump.

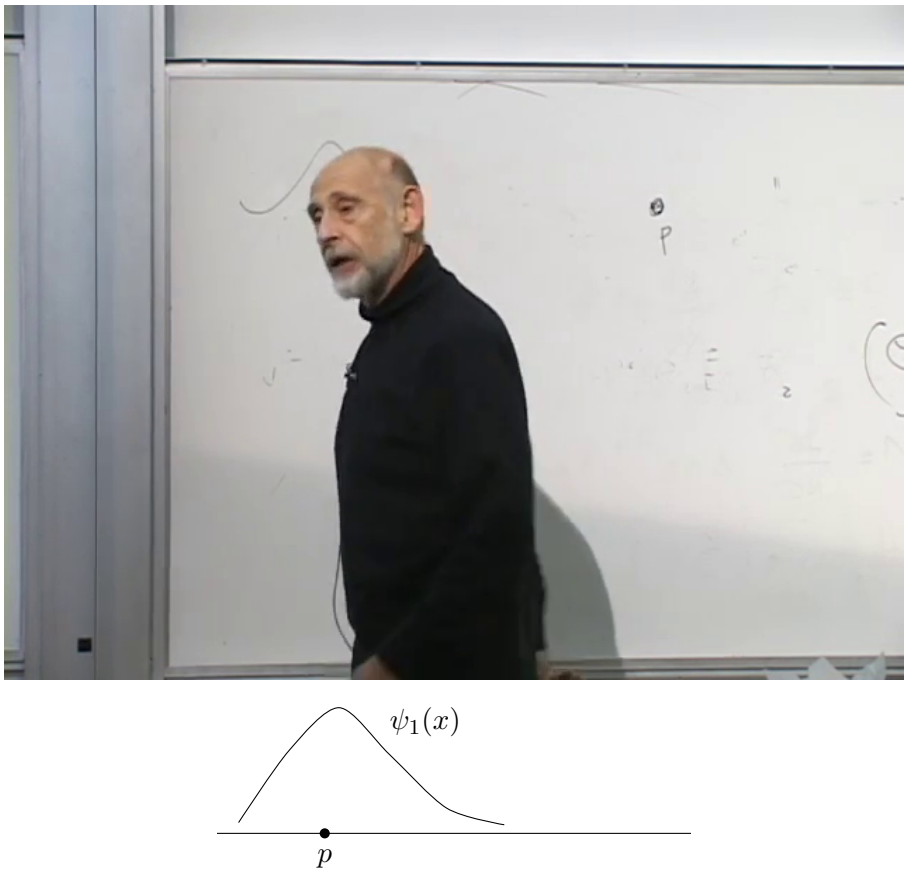
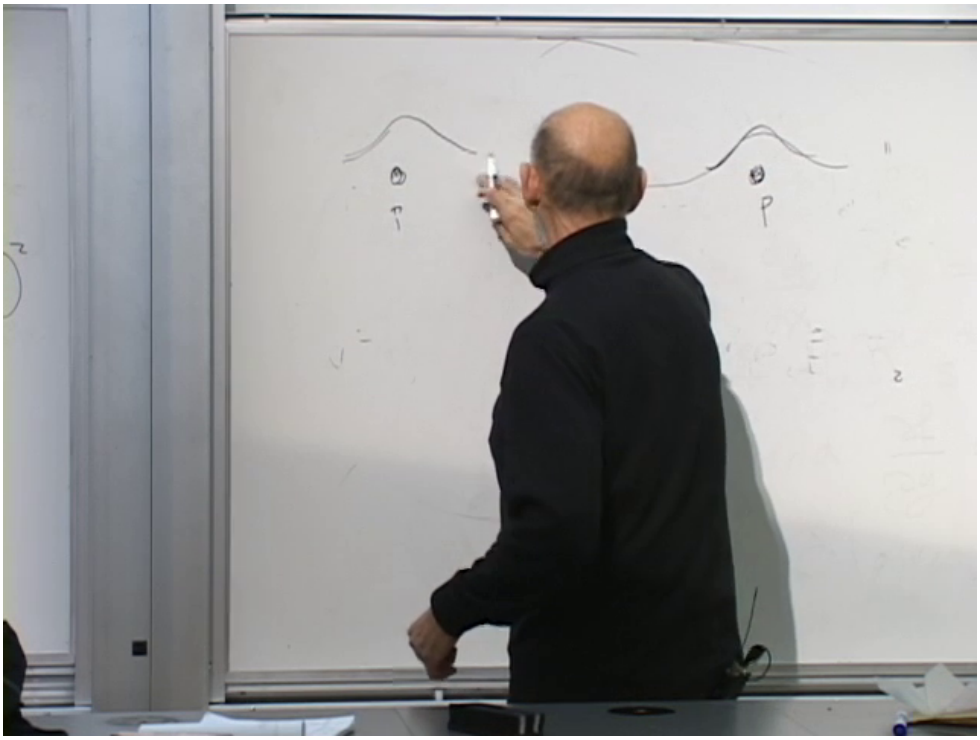


Figure 1.1: Single hydrogenic wave-function sketch. The frame documents the board layout; the TikZ curve is a cleaned schematic reading rather than a literal transcription of notation.

The faint point label beneath the chalk mark looks like a handwritten P, but it is too uncertain to be used as reliable notation. What matters is the shape: a bound-state profile localized near one force center.

Now place a second proton at finite but large separation. By symmetry there is a second localized state of the same energy, centered near the right proton.

At infinite separation the two states are degenerate. The electron may be localized near the left proton or near the right proton, and the energy is the same.



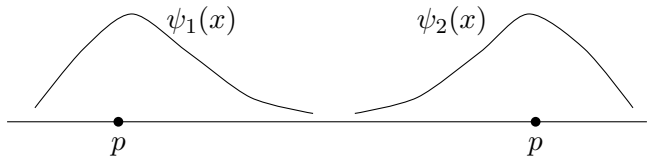


Figure 1.2: Two localized electron wave functions. The lecture now moves from these two one-center states to an equilibrium superposition.

1.3.1 Question & Answer

Question. Why is the two-proton system not exactly degenerate once the separation is finite?

Answer. Because once the protons are not infinitely far apart, processes become possible that were impossible before. In particular, the electron can tunnel from one center to the other. The lecture first describes this qualitatively, but the minimal mathematical reconstruction is a two-state Hamiltonian. If $|L\rangle$ and $|R\rangle$ denote the left- and right-localized states, then finite separation introduces off-diagonal matrix elements:

$$H|L\rangle = E_0|L\rangle - t|R\rangle, \tag{1.8}$$
$$H|R\rangle = E_0|R\rangle - t|L\rangle, \tag{1.9}$$

with $t > 0$ a tunneling amplitude.

The eigenstates are therefore the symmetric and antisymmetric combinations

$$|+\rangle = \frac{|L\rangle + |R\rangle}{\sqrt{2}}, \qquad E_+ = E_0 - t, \tag{1.10}$$

$$|-\rangle = \frac{|L\rangle - |R\rangle}{\sqrt{2}}, \qquad E_- = E_0 + t. \tag{1.11}$$

Translated back into wave functions, the lower equilibrium state is approximately

$$\psi_+(x) \approx \frac{\psi_1(x) + \psi_2(x)}{\sqrt{2}}. \tag{1.12}$$

The word "approximately" matters. The lecture is explicit that the true equilibrium wave function is not *exactly* the naive sum. In the overlap region between the two wells it is slightly different. The accepted frame does not settle the exact contour there, and in any case Susskind himself says he did not draw it very well. But the physical point is stable: once tunneling is allowed, the true equilibrium state has lower energy than the two isolated one-center states.

The lecture also says what one should not do. One should not picture this as a literal observed trajectory, with the electron watched in transit. As in the rest of quantum mechanics, trying to watch the tunneling ruins the phenomenon. The exchange picture is a way of describing the state, not a classical movie of a particle moving through space.

1.4 From Lowered Energy to an Effective Force

Now the lecture shifts, very deliberately, from the shape of the state to the shape of the energy. At large proton separation the electron energy is just the familiar one-center ground-state energy. At smaller separation, once tunneling and superposition are allowed, the equilibrium energy is slightly lower. Therefore the energy of the whole two-proton one-electron system becomes a function of the proton separation r.

This is the exact point at which the molecular example reconnects with the field-theory example. We may regard that separation-dependent energy as an effective potential $V(r)$. The board then adds the second, lower sketch: a qualitative graph of energy against separation.

The force is now read from the energy gradient:

$$F(r) = -\frac{dV}{dr}. \tag{1.13}$$

The lecture phrases it in plain language: force pushes toward decreased energy. If the energy drops as the protons move together, then the system experiences an effective attraction.

This is the molecular version of the same logic we saw earlier for classical fields. In one case the energy change comes from the overlap of two electric fields. In the other it comes from the lowering of the quantum-mechanical equilibrium energy in a two-center problem. In both cases, force comes from a configuration-dependent energy.

Several audience questions sharpen the intended picture, and they are worth preserving because otherwise the discussion becomes too loose.

What the picture is not.

- It is not mainly a screening story. The essential point is not that the electron merely reduces the proton charge seen by the other proton.
- It is not a third independent force in addition to the familiar ones. Rather, the Schrödinger problem in the presence of two force centers has a lower equilibrium energy than the sum of two isolated one-center problems.
- It is not best described as ordinary two-particle entanglement. The lecture says more pointedly that the relevant structure is the presence of off-diagonal elements in the Hamiltonian.
- It is not a literal watched trajectory. The worldline picture is schematic, useful for intuition and dangerous if taken too seriously.

The lecture also emphasizes a limitation. In the one-electron system the two protons still carry net positive charge, so Coulomb repulsion remains. The exchange attraction is short-ranged; the electrostatic repulsion is longer-ranged. Thus attraction does not automatically mean stable binding at all separations. In a neutral two-electron system, by contrast, the long-range Coulomb piece is much less dominant, and the same exchange mechanism shows up more cleanly as ordinary molecular binding.

The exchange picture can nevertheless be drawn, and the lecture does draw it in words.

1.5 Photon Exchange and the First Entries of the Table

Now that the exchange picture has been made visible in a molecular setting, the lecture returns to electrodynamics. Can the Coulomb force be told in the same language? Yes. A charged particle interacts with the electromagnetic field, and the quanta of that field are photons. Classically one solves for an electric field around the charge. Quantum mechanically one thinks instead in terms of emission and absorption of photons.

The lecture makes this more concrete. A physical charged particle is not just a bare charge. It is a quantum-mechanical superposition of states containing no photons, one emitted photon, two photons, and so on, with photons continually emitted and reabsorbed. If another charged particle is nearby, then a photon emitted by one may be absorbed by the other. That shared-photon process changes the energy of the two-particle system away from the simple sum of two isolated energies. That energy shift is the Coulomb interaction in particle language.

This is the place where Susskind explicitly closes the triangle. Particles, fields, and forces are now in one-to-one-to-one correspondence. When people say there are only four forces in nature, the lecture pushes back. In the exchange sense, every exchangeable quantum can mediate a force in some appropriate context. Electrons do so in molecular physics. Photons do so in electrodynamics.

Only after this bridge has been built does the lecture finally allow itself to start the table of particles. And even here the style matters. He does not want to dump the whole Standard Model on the board at once. Better to proceed in small groups, because the full thing is, as he says, a monstrosity.

The accepted frame should remain visible, because it captures exactly that switch in board logic.

The board evidence supports the secure identification

$$\gamma \leftrightarrow A. \tag{1.14}$$

The lecture then explains in words that A is the electromagnetic vector potential. In a later, more covariant notation one would often write A_μ, but the board itself only visibly shows A, so that is the notation we keep at first pass.

The table columns are introduced in a specific order: name, symbol, type, charge, baryon number, and mass. Here "type" means simply boson or fermion. The first secure row is the photon row; the next row, for the electron and positron, comes from the transcript rather than from the crop itself.

Name	Symbol	Field	Type	Q	B	Mass
Photon	γ	A	boson	0	0	0
Electron/positron	$e^\pm$	ψ_e	fermion	∓ 1	0	0.51 MeV

Table 1.1: A cleaned continuation of the lecture's first table entries. Only the photon row is directly visible in the accepted crop; the electron row is added from the transcript.

Here the lecture slows down again for units and conventions. Electric charge is measured in units of the electron charge. Thus we simply write

$$Q(e^-) = -1, \qquad Q(e^+) = +1. \tag{1.15}$$

Likewise mass is measured in energy units, since $E = mc^2$. The historically natural unit is the electron volt, and in particle physics we quickly move to MeV and GeV. The electron mass is

$$m_e \approx 0.51\,\text{MeV}. \tag{1.16}$$

The lecture also remarks that it is conventional to speak of the electron as the particle and the positron as its antiparticle, though the relation is actually mutual; choosing which one to call "the particle" is a convention of language.

The elementary QED process is then introduced: one electron in, one electron out, one photon attached. By rearranging the legs of that one vertex one reads the same diagram as emission, absorption, or positron emission. At the lecture's level the interaction term is described schematically as a product of one electron field in, one electron field out, and one photon field:

$$\mathcal{L}_{\text{int}} \sim e\,\psi_e^\dagger \psi_e A. \tag{1.17}$$

This is intentionally schematic. A more modern covariant form would refine both the spinor structure and the photon field notation, but the lecture's real point is simpler:

$$\mathcal{A}(e \to e + \gamma) \propto e, \qquad P(e \to e + \gamma) \propto e^2. \tag{1.18}$$

A schematic vertex is therefore useful.

And the force between charges is the many-body consequence of such elementary emission and absorption events:

At this point the opening promise can finally be kept. We are ready to begin naming the players.

1.6 Quarks, Baryon Number, and the Puzzle of Families

The next move is historically important. In an earlier era one might have gone directly from electrons and photons to protons and neutrons. The lecture instead says: no, modern particle physics places quarks first. Protons and neutrons are to be understood, at least naively, as composites of quarks.

There are six quark flavors,

$$u,\ d,\ s,\ c,\ b,\ t, \tag{1.19}$$

and the lecture is almost amused by the naming conventions. Up and down are not really up and down. Strange is not stranger than the others. Charm, bottom, and top are historical accidents. The names are not the structure.

The structure comes in two parts. First, all quarks are fermions. Second, their charges repeat in pairs:

$$Q(u) = Q(c) = Q(t) = +\frac{2}{3}, \qquad Q(d) = Q(s) = Q(b) = -\frac{1}{3}. \tag{1.20}$$

Thus the trio (d, u), (s, c), and (b, t) exhibits a pattern of replicated charge assignments. But the masses do not remotely repeat:

$$m_u \sim 5\,\text{MeV}, \qquad m_d \sim 10\,\text{MeV}, \tag{1.21}$$

$$m_s \sim 100\,\text{MeV}, \qquad m_c \sim 10^3\,\text{MeV}, \tag{1.22}$$

$$m_b \sim 5\,\text{GeV}, \qquad m_t \sim 170\,\text{GeV}. \tag{1.23}$$

So the lecture presses on a tension that is central to the Standard Model. The quarks are in one sense copies of one another; in another sense their mass scales wander across several orders of magnitude. That repetition of families without any satisfying explanation is one of the deep puzzles of the subject.

The lecture also restores some phenomenological perspective. Of the six quark flavors, only u and d appear abundantly in ordinary matter. The heavier ones decay down the chain. So low-energy physics is dominated by the first family, while the others are present as heavier echoes.

1.6.1 Question & Answer

Question. Why does a quark carry baryon number 1/3, and can baryon number be violated?

Answer. Baryon number is introduced operationally, not metaphysically. One first counts protons and neutrons. By convention,

$$B(p) = 1, \qquad B(n) = 1. \tag{1.24}$$

That normalization could have been chosen differently; the lecture explicitly remarks that one could have called it 2, or 7, or π, as long as it scaled consistently. Once the proton is recognized as a three-quark object, however, consistency forces

$$B(q) = \frac{1}{3}, \qquad B(\bar{q}) = -\frac{1}{3}. \tag{1.25}$$

So the factor 1/3 is not a new mystery. It is the inherited normalization of a bookkeeping convention.

The sharper question is whether baryon number is exact. Empirically it is conserved to extremely high precision, but the lecture refuses to promote that fact into an exact law. A decay such as

$$p \to e^+ + \gamma \tag{1.26}$$

would conserve electric charge and still violate baryon number. No such process has been observed; the proton lifetime must be at least of order 10^{33} to 10^{34} years. But that does not amount to a proof that baryon number is absolute. The lecture leaves the question open.

A further caution is also preserved. Quarks are not observed as isolated free particles. One can certainly detect evidence of quarks, but not a lone quark sitting in isolation waiting to be measured.

1.7 From Quarks to Hadrons: Nucleons, Strange Baryons, and Mesons

Once the quark charges are known, familiar hadrons can be rebuilt. The first examples are the neutron and proton:

$$n = ddu, \qquad p = uud. \tag{1.27}$$

Worked example. The lecture explicitly checks the charges:

$$Q(n) = Q(d) + Q(d) + Q(u) = -\frac{1}{3} - \frac{1}{3} + \frac{2}{3} = 0, \tag{1.28}$$

$$Q(p) = Q(u) + Q(u) + Q(d) = \frac{2}{3} + \frac{2}{3} - \frac{1}{3} = 1. \tag{1.29}$$

Now comes a characteristic Susskind move. Forget photons for a moment. If we ignore electromagnetism and focus only on the quark content, then proton and neutron look almost like the same object with u and d interchanged. This suggests a near-symmetry. It is not exact, because u and d do not have exactly the same masses.

That near-symmetry is good enough to tell us the sign of the proton-neutron mass difference. Since $m_d > m_u$, the neutron, which contains more down quarks, should be slightly heavier. And it is. Electromagnetic self-energy by itself would push the other way and make the proton heavier, because the proton is charged. The observed ordering therefore tells us that the $d - u$ mass difference wins over the electromagnetic correction.

Remark 1.1. The lecture also points out the famous mismatch in the nucleon mass budget. The quark masses add up to only a few MeV, whereas the proton and neutron masses are around 940 MeV. The missing contribution is not mysterious "stuff" hidden in the nucleon; it is energy carried by the gluonic and binding dynamics. The lecture's summary is blunt: masses do not simply add relativistically for bound systems.

The same constituent logic produces strange baryons. Since s and d have the same electric charge, one may replace a down quark by a strange quark without changing the charge pattern:

$$uds, \qquad uss, \qquad uus. \tag{1.30}$$

These are analogs of the ordinary nucleons, but heavier, and therefore unstable. They decay down to lighter baryons plus other particles. The lecture is not yet trying to classify the decays in detail; it is simply establishing that the strange quark may be inserted into the same three-quark structures.

At this point the lecture inserts a useful rule of thumb. Why do three quarks bind into baryons? Why not two quarks? In the early days of quark theory this was a genuine puzzle. At the level of this lecture no full derivation is offered. Instead one adopts the empirical rule that the bound states one actually encounters carry charges that are integer multiples of the electron charge. Baryons do so with three quarks; mesons do so with quark-antiquark pairs. The lecture explicitly treats this as a working rule before deeper explanations are taken up.

Mesons are therefore introduced next as $q\bar{q}$ composites. Their baryon number vanishes:

$$B(q\bar{q}) = \frac{1}{3} - \frac{1}{3} = 0. \tag{1.31}$$

The first mesons to discuss are the ones built only from u and d, because those are the light quarks: they are abundant, comparatively stable, and cheap to produce.

The natural flavor basis is

$$|\bar{u}d\rangle, \qquad |u\bar{d}\rangle, \qquad |\bar{u}u\rangle, \qquad |\bar{d}d\rangle. \tag{1.32}$$

The charged states are immediate:

$$|\pi^-\rangle = |\bar{u}d\rangle, \qquad |\pi^+\rangle = |u\bar{d}\rangle. \tag{1.33}$$

1.7.1 Question & Answer

Question. Why are the neutral meson states written as superpositions rather than as separate $\bar{u}u$ and $\bar{d}d$ particles?

Answer. Because at this point the lecture is explicitly doing quantum mechanics. The states $|\bar{u}u\rangle$ and $|\bar{d}d\rangle$ are a basis, not automatically the physical particles. One may superpose them, and in fact the lecture says that neither one by itself corresponds to a distinct particle. The two orthogonal combinations are

$$|\pi^0\rangle \approx \frac{|\bar{u}u\rangle - |\bar{d}d\rangle}{\sqrt{2}}, \tag{1.34}$$

$$|\eta\rangle \approx \frac{|\bar{u}u\rangle + |\bar{d}d\rangle}{\sqrt{2}}. \tag{1.35}$$

This is also the lecture's first glimpse of isospin. Upness and downness are treated as a two-state internal degree of freedom, analogous in a loose way to spin up and spin down. In that language the neutral combinations are not arbitrary. They are the natural sum and difference states in the two-dimensional flavor space.

The pion triplet π^+, π^0, π^- goes together because the three masses are close. In the limit $m_u = m_d$, they would be even closer, ideally degenerate. The orthogonal neutral state is the η, which behaves similarly in many respects but sits higher in mass:

$$m_\pi \approx 140\,\text{MeV}, \qquad m_\eta \approx 500\,\text{MeV}. \tag{1.36}$$

The lecture treats the anomalous heaviness of the η as an important but more complicated story, not to be unpacked in this first chapter.

A further empirical remark from the lecture is worth keeping: the pions are comparatively long-lived because they do not decay by the strong interaction. Their decays proceed only through weaker channels.

Finally, once the strange quark is admitted, one obtains the strange mesons or K-mesons by replacing d or $\bar{d}$ with s or $\bar{s}$. The flavor states are

$$\bar{u}s, \qquad u\bar{s}, \qquad \bar{d}s, \qquad d\bar{s}. \tag{1.37}$$

Their charge assignments are

$$Q(\bar{u}s) = -1, \qquad Q(u\bar{s}) = +1, \tag{1.38}$$

$$Q(\bar{d}s) = 0, \qquad Q(d\bar{s}) = 0. \tag{1.39}$$

So there are two charged K-mesons and two neutral ones.

Counting all light u, d, s quarks and antiquarks gives

$$3 \times 3 = 9 \tag{1.40}$$

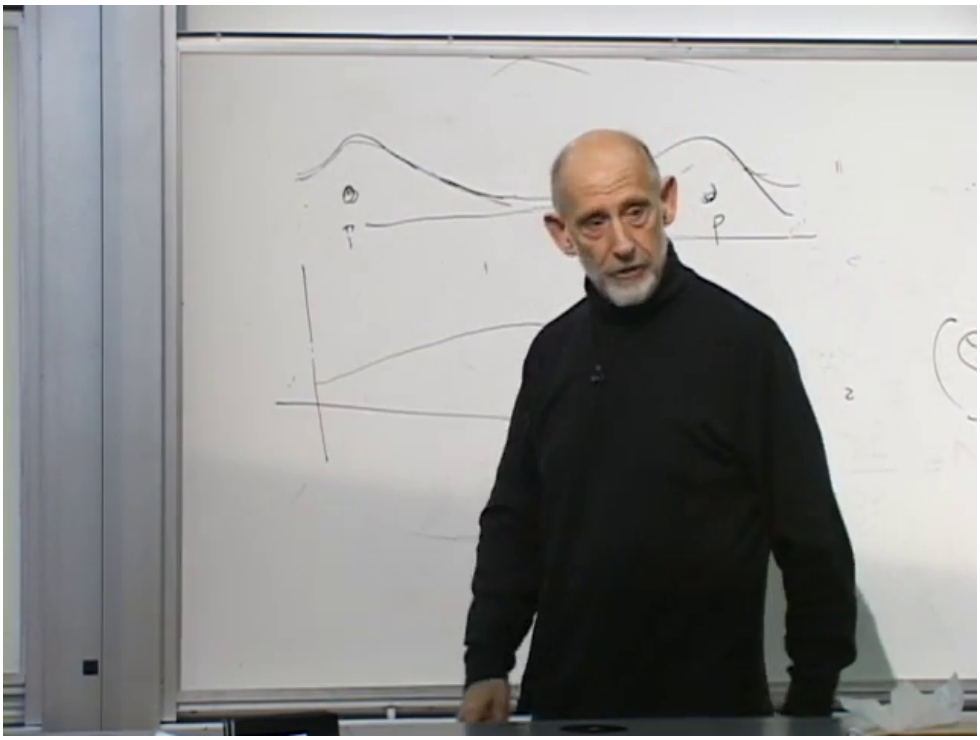
light flavor combinations in all. The lecture stops there, deliberately before complete classification. The first map of the territory has been drawn, but only in outline. That is exactly the right stopping point for Lecture 1.

1.8 Summary

The lecture begins by postponing the particle list and ends by justifying that postponement. We first see, in classical electrodynamics, that force can be read from the part of the field energy that depends on configuration. We then see the same idea retold in a different language through molecular exchange: two centers, tunneling, superposition, lowered energy, and therefore attraction.

Only after that bridge is built does the lecture shift into catalogue mode. The photon and its field A, the electron and its field ψ_e, the QED vertex, then quarks, baryon number, replicated families, nucleons, strange baryons, and neutral meson superpositions. The Standard Model is not presented as a finished elegant structure. It is presented, more honestly, as a partly organized and partly mysterious collection of particles whose patterns are visible long before their deepest explanation is in hand.

That is the right tone for the beginning of the subject. The lecture does not merely list what exists. It shows us how the concepts fit together well enough that the list can begin to mean something.



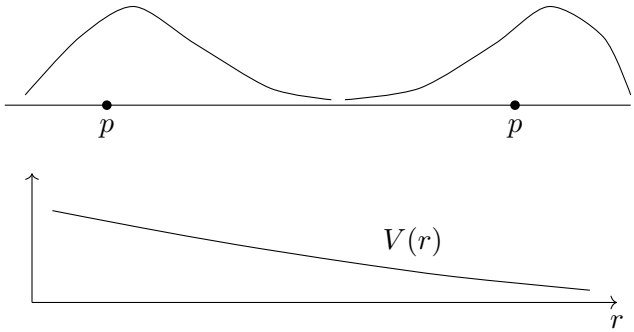


Figure 1.3: Wave-function overlap and effective proton potential. The board image supports the two-level layout; the label $V(r)$ is a cleaned choice, since the frame itself does not fix whether the intended ordinate was E or V.

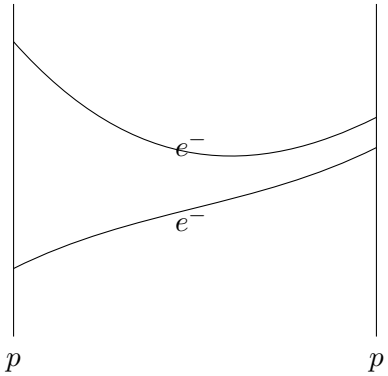


Figure 1.4: Transcript-backed schematic of electron exchange between two proton worldlines. This is an explanatory reconstruction of the lecture's picture, not a direct board transcription.

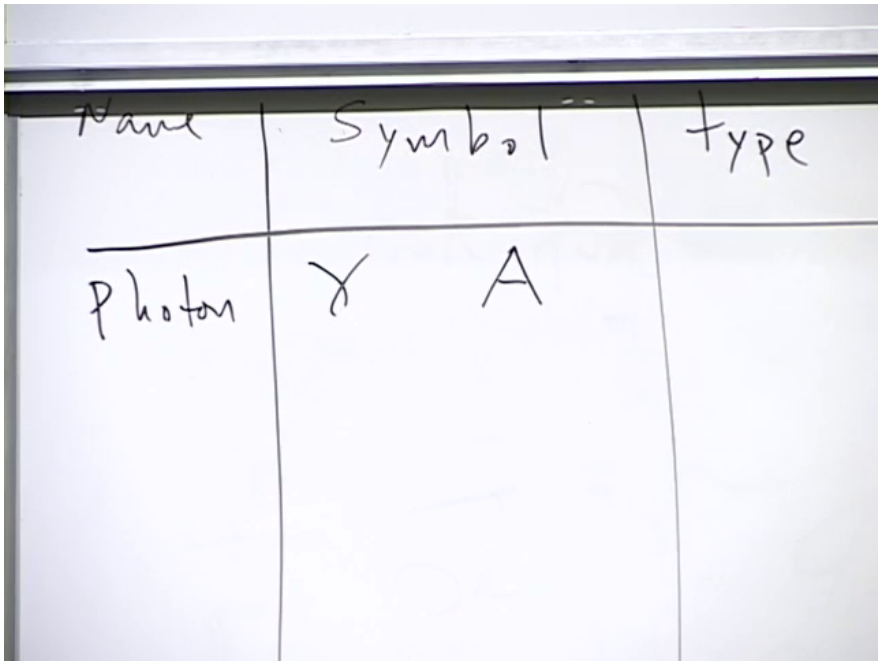


Figure 1.5: Photon symbol and vector-potential notation. The crop securely shows the photon row and the entries γ and A.

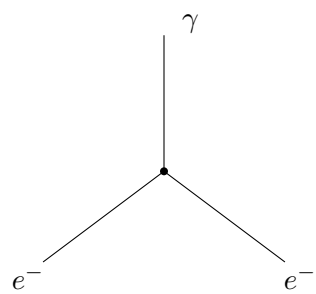


Figure 1.6: Schematic QED vertex: one electron in, one electron out, one photon attached. By crossing the legs, the lecture reads the same vertex in several ways.

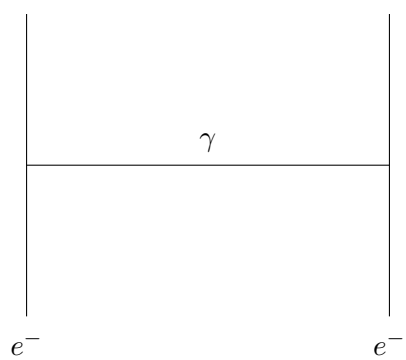


Figure 1.7: Transcript-backed schematic of photon exchange. In the lecture this is the quantum-field-theoretic retelling of the Coulomb interaction.

Chapter 2

Lecture 2: New Revolutions in Particle Physics: Standard Model

These notes follow Leonard Susskind's second lecture in the supplementary Standard Model sequence, with curation by LazyingArt LLC. The lecture begins with a warning and a promise. Quantum chromodynamics is far too large a subject for a single meeting, so we will not attempt its deep formal mathematics here. What we will do instead is extract the basic picture. The path is deliberate: QCD is first framed by analogy with QED, then the lecture backs up into the mathematics of spin, and only after that uses the same machinery to introduce isospin, the Δ puzzle, color, gluons, and the first qualitative form of QCD.

2.1 QCD by Analogy, and Why We Must Revisit Spin

Susskind opens by placing quantum chromodynamics beside quantum electrodynamics. QED is the theory of electrons and photons; in the atomic setting it is also the theory of the Coulomb force, the field that binds electrons to nuclei. In the same sense, QCD is the theory of quarks and gluons, and of the composite things that quarks and gluons make. Those composites are hadrons. Protons and neutrons are baryons, with net baryon number; mesons are quark–antiquark pairs; and in either case the gluons are part of the binding medium. The lecture wants us to see that QCD is not merely a list of particles. It is a theory of constituents and of the force that holds them together.

But before QCD proper begins, the lecture steps backward. Why? Because the mathematics of spin will come back twice: first almost directly as isospin, and then again, in a looser but still structurally related way, as color. The lecture explicitly refuses a group-theory treatment. We are told that group theory is really lurking underneath, but the working language will be angular momentum, because that language is already familiar and already rich enough for the combinatorics we need.

2.2 Angular Momentum Multiplets and Two Spin-Half Particles

The reminder begins with the commutator algebra of angular momentum:

$$[L_x, L_y] = i\hbar L_z, \tag{2.1}$$

together with the cyclic companions

$$[L_y, L_z] = i\hbar L_x, \qquad [L_z, L_x] = i\hbar L_y. \tag{2.2}$$

The lecture immediately emphasizes what this means operationally. The physics is rotationally symmetric, but our description does not have to display that symmetry at every step. We choose a preferred axis, conventionally the z-axis, and we ask for the component of angular momentum along that axis. This is not symmetry breaking in nature; it is only the expedient choice that makes the representation manageable. Since the components do not commute, we cannot measure them all at once. So we focus on one of them.

From this point of view a spin multiplet is organized by the largest value attained by the z-component. The lecture says L aloud, while the blackboard uses ℓ; to avoid confusion with the operators L_x, L_y, L_z, we will write the multiplet label as ℓ. The familiar counting rule is then

$$m_z = \ell, \ell - 1, \ldots, -\ell, \qquad \#\text{ of independent states} = 2\ell + 1. \tag{2.3}$$

That is the key fact the lecture wants us to carry forward.

Now let us put together two spin-half particles, ignoring orbital angular momentum entirely. The product basis is

$$|\uparrow\uparrow\rangle, \qquad |\uparrow\downarrow\rangle, \qquad |\downarrow\uparrow\rangle, \qquad |\downarrow\downarrow\rangle. \tag{2.4}$$

The maximum m_z is $+1$, attained when both spins point up, and the minimum is -1, attained when both point down. But there is already something to notice: there are two distinct basis states with $m_z = 0$, namely $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$. This is precisely where the lecture stops being a review and becomes a problem.

The board frame is useful not only because the mixed-spin sum has been circled, but also because the earlier spin algebra still remains visible at the top of the board. One can literally see the lecture passing from the commutator and the $2\ell+1$ counting rule to the first nontrivial example. In cleaned notation the circled state is

$$\frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}. \tag{2.5}$$

The factor $1/\sqrt{2}$ is the standard normalized completion. It is not yet visible in the earlier frame, but the transcript makes clear that this is the next move in the lecture.

2.2.1 Question & Answer

Question. If two spin-half particles give four basis states, why do they not form a single spin-one multiplet?

Answer. Because a spin-one multiplet has only three states:

$$m_z = +1, \quad 0, \quad -1.$$

The two aligned states already account for the $m_z = \pm 1$ members. So a spin-one multiplet can contain only one linear combination from the two-dimensional $m_z = 0$ sector. There is still one orthogonal combination left over. It cannot be spin two, because that would require five states altogether. It cannot be anything half-integral, because the total spin of two spin-half particles is integral. The only remaining possibility is spin zero.

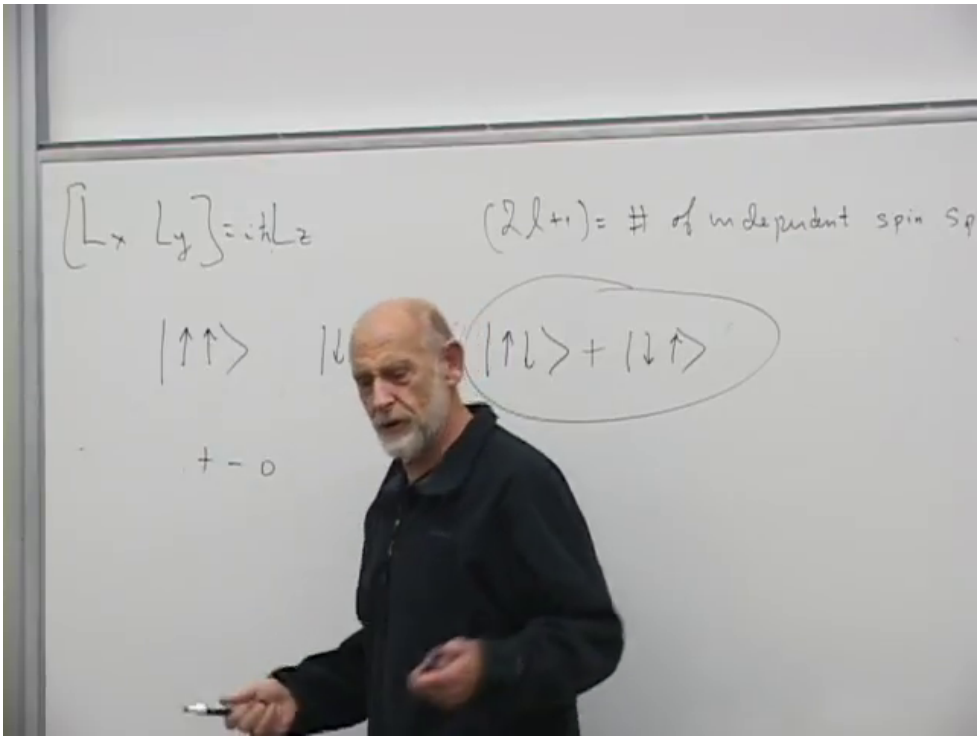


Figure 2.1: Symmetric mixed-spin state highlighted on the board. The frame preserves the moment at which the mixed-spin sum is singled out before the normalization is written explicitly.

2.3 Triplet, Singlet, and the Meaning of the Symmetric State

The lecture now resolves the counting puzzle exactly in that way. We diagonalize the $m_z = 0$ subspace into symmetric and antisymmetric combinations, and the four product states reorganize themselves as

$$\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0, \tag{2.6}$$

$$\ell = 1: \qquad |\uparrow\uparrow\rangle, \qquad \frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}, \qquad |\downarrow\downarrow\rangle, \tag{2.7}$$

$$\ell = 0: \qquad \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}. \tag{2.8}$$

This is a familiar identity, but in the lecture it is not merely quoted. It is forced by counting, and then interpreted physically.

The essential interpretive point is the one Susskind presses hardest: the symmetric mixed-spin state has $m_z = 0$, but it is not spin zero. It belongs to the $\ell = 1$ triplet. That is why it rotates into the $m_z = \pm 1$ states when the axis is changed. Zero along the chosen axis is not the same thing as zero total angular momentum. Only the antisymmetric combination has total angular momentum zero:

$$\frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}} \quad \longleftrightarrow \quad \ell = 0. \tag{2.9}$$

The lecture then sharpens the distinction with a physical test. Suppose we ignore every conservation law except angular momentum. Which of these states could simply disappear? The aligned states

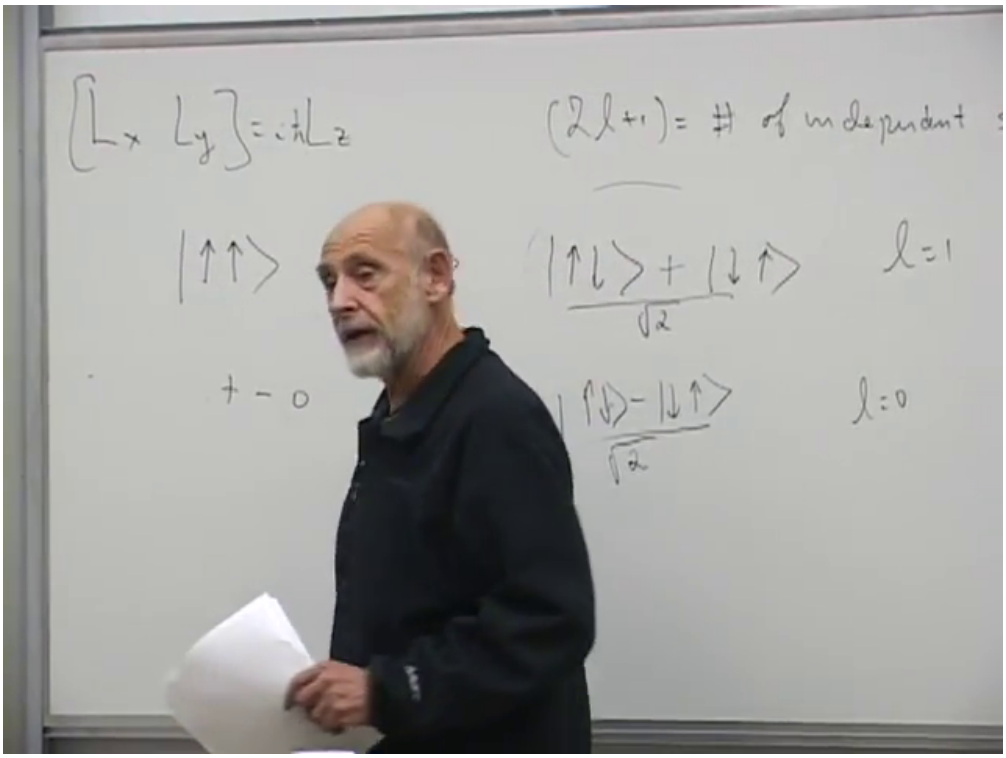


Figure 2.2: Triplet and singlet decomposition of two spin-$\frac{1}{2}$ states. The board layout itself matters: the three $\ell = 1$ states are grouped together, while the antisymmetric $\ell = 0$ state is set apart below.

clearly cannot. The symmetric $m_z = 0$ state also cannot, because although its z-component vanishes, it still belongs to the spin-one multiplet. The only combination that could disappear on angular-momentum grounds alone is the antisymmetric singlet. In that sense the singlet behaves, as Susskind says, like empty space as far as angular momentum is concerned.

This also clarifies a question from the room. If we measure the symmetric mixed-spin state along the z-axis, we obtain zero. But rotate the axis, and the same state reveals the nonzero angular momentum appropriate to the triplet. The lecture wants us to keep that distinction in mind, because exactly this sort of counting-with-interpretation will reappear when spin is replaced by isospin.

2.4 Three Spins as the Template for Isospin

At this point the lecture pauses and asks, in effect, why we are doing all this. Before answering, it runs the same counting argument once more for three spin-half particles. There are

$$2 \times 2 \times 2 = 8 \tag{2.10}$$

product states altogether. If all three spins point up, the maximum m_z is $+\frac{3}{2}$, so a spin-$\frac{3}{2}$ multiplet must be present. That multiplet contains

$$2\left(\frac{3}{2}\right) + 1 = 4 \tag{2.11}$$

states. Four states therefore remain. They cannot form another spin-$\frac{3}{2}$ multiplet, because the $\pm\frac{3}{2}$ extrema have already been used. So the only possibility is spin-$\frac{1}{2}$, and since four states remain,

spin-$\frac{1}{2}$ appears twice:

$$\frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2} \oplus \frac{1}{2}. \tag{2.12}$$

The lecture's phrase is: let us keep that in mind.

Now the reason for the spin review arrives. What Susskind really wants to discuss is isospin, the first important internal analogue of spin in particle physics. It does not refer to the rotation of physical space. It refers to a two-state internal structure, mathematically isomorphic to the spin-half system. The secure point, especially where the transcript becomes noisy, is that isospin is an internal copy of the same mathematics, not a literal spatial spin.

The justification is physical, not merely formal. The natural scale of hadron physics is a few hundred MeV. On that scale the up and down quarks are both very light, and, more importantly, nearly degenerate. Electromagnetic effects also distort the symmetry only slightly in the hadronic setting we are considering. So as a first approximation we may neglect the mass difference between u and d and ignore their electric charges. Then the pair

$$u, \qquad d$$

behaves as a doublet under an approximate internal symmetry. In that approximation we may treat the up and down quarks as the analogue of spin up and spin down.

Historically isospin predates quarks; the lecture briefly recalls that it goes back to Heisenberg and to the older interchange of proton and neutron. But here it is introduced in the modern order: the light quark pair provides the internal two-state system, and from there the old isospin language becomes natural again.

2.5 From Spin to Isospin: Up and Down Quarks, Proton and Neutron

Once the u–d doublet is in place, the next step is almost forced. A single quark is not the simplest object we observe in the laboratory. The simplest hadrons in the discussion are three-quark states, namely the proton and the neutron. By direct analogy with the three-spin problem, three light quarks can make an isospin-$\frac{1}{2}$ sector and an isospin-$\frac{3}{2}$ sector. The lecture chooses to begin with the milder case, the isospin-$\frac{1}{2}$ doublet.

The two states are

$$p \sim uud, \qquad n \sim udd. \tag{2.13}$$

This is the hadronic counterpart of the spin-half multiplet. Two states only, related by the interchange $u \leftrightarrow d$.

Their charges follow from

$$Q(u) = +\frac{2}{3}, \qquad Q(d) = -\frac{1}{3}. \tag{2.14}$$

Thus

$$Q(p) = \frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +1, \qquad Q(n) = \frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0.$$

The small mass difference between proton and neutron is one of the lecture's practical signs that the symmetry is only approximate, but still good enough to organize the strong-interaction multiplets.

At this point the lecture briefly becomes more precise than it wants to remain. The quarks are labeled by positions $1, 2, 3$, and the board writes an antisymmetrized factor involving the two up quarks:

$$d_1(u_2u_3 - u_3u_2). \tag{2.15}$$

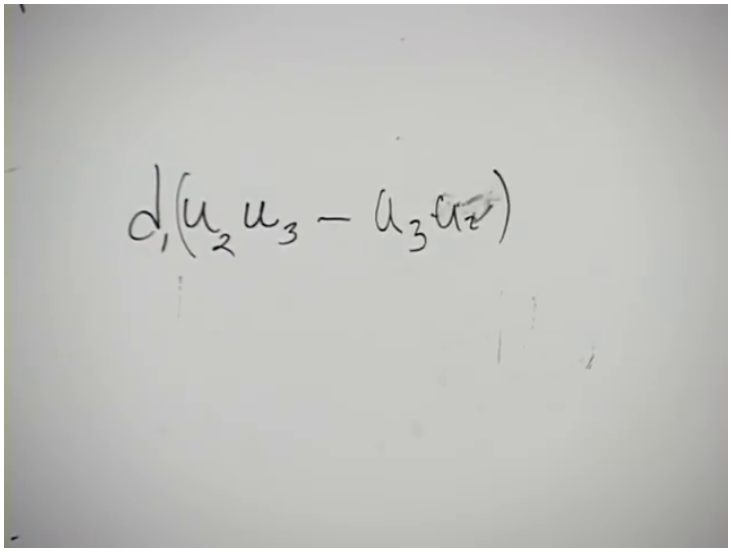


Figure 2.3: Antisymmetrized two-u-quark factor in the proton state. This is documentary evidence for the board expression only; it is not the full proton wavefunction.

The lecture immediately tells us not to lean too hard on this formal step. It is being precise, but only locally. The board shows one antisymmetrized factor, not the full proton wavefunction. What matters is the analogy with the two-spin singlet: an antisymmetric pair behaves like the zero-angular-momentum combination in the earlier discussion, and together with the third quark one arrives at an overall isospin-$\frac{1}{2}$ state. Susskind then says, in effect, never mind the fancy part; the important point is that the proton and neutron form the isospin-half doublet.

If one wished to normalize the displayed two-u-quark factor in the usual way, one would insert a factor of $1/\sqrt{2}$, but that factor is not written on the board and is not needed here. The chapter should preserve the actual board evidence and no more.

2.6 The Delta Multiplet and the Pauli Puzzle

Now the lecture shifts to the stronger case. If three light quarks can make isospin $\frac{1}{2}$, they can also make isospin $\frac{3}{2}$. The analogue of the spin-$\frac{3}{2}$ quartet is the Δ multiplet:

$$\Delta^{++} \sim uuu, \qquad \Delta^{+} \sim uud, \qquad \Delta^{0} \sim udd, \qquad \Delta^{-} \sim ddd. \tag{2.16}$$

All four states are related by the approximate isospin symmetry and therefore have nearly the same mass, up to the small effects of the u–d mass difference.

At the same time these states are not just the proton and neutron in disguise. Their ordinary spin is different. The proton and neutron have actual spin $\frac{1}{2}$; the Δ states have spin $\frac{3}{2}$. In the lecture this means that the three ordinary spins are aligned, and the three isospins are aligned as well. The

quartet therefore makes very concrete the combinatorial pattern that we first met in the abstract three-spin problem.

Their charges are

$$Q(\Delta^{++}) = +2, \qquad Q(\Delta^{+}) = +1, \qquad Q(\Delta^{0}) = 0, \qquad Q(\Delta^{-}) = -1. \tag{2.17}$$

The lecture quotes the proton and neutron mass scale as roughly 940 MeV and the Δ mass scale as roughly 1200 MeV. That is enough to make the Δ unstable.

A worked example is the decay of the doubly charged member. Start from

$$\Delta^{++} \sim uuu. \tag{2.18}$$

As the quarks pull apart, a quark–antiquark pair may appear. The lecture first toys with a $u\bar{u}$ insertion, then rejects it on energetic grounds and chooses $d\bar{d}$. That gives

$$uuu \longrightarrow uud + u\bar{d} \tag{2.19}$$
$$= p + \pi^{+}. \tag{2.20}$$

Charge conservation is immediate:

$$Q(p) = +1, \qquad Q(\pi^{+}) = Q(u) + Q(\bar{d}) = \frac{2}{3} + \frac{1}{3} = +1.$$

So the final state again has total charge +2. The mass balance is also consistent with the lecture's rough numbers:

$$1200 \text{ MeV} > 940 \text{ MeV} + 140 \text{ MeV},$$

so the decay has phase space available. This is why the Δ states are real but very short-lived resonances.

2.6.1 Question & Answer

Question. How can a Δ^{++} made of three up quarks avoid violating Fermi statistics?

Answer. This is the central tension of the lecture. In Δ^{++} the three quarks are all u, and because the baryon has spin $\frac{3}{2}$, their ordinary spins are aligned as well. So if spin and flavor were the whole story, we would seem to have three identical fermions in the same state. That is forbidden by the Pauli principle. The lecture presents this not as a minor technical issue but as a historical contradiction that had to be resolved. The resolution is that quarks must carry one more label, hidden at first from view, so that the three quarks in the Δ^{++} are not identical in every respect. That extra label is color.

The lecture even pauses to note why the Δ was so important historically. In the proton and neutron one could imagine the problem being disguised by more complicated symmetrization. In the Δ^{++} the contradiction is blunt: three up quarks, parallel spin, parallel isospin, and something is plainly wrong unless a new quantum number is introduced.

2.7 Color, Gluons, and the Minimal Picture of QCD

Color now enters as the hidden label demanded by the Δ puzzle. The lecture uses the conventional names

$$r, \qquad g, \qquad b,$$

with corresponding anti-colors

$$\bar{r}, \qquad \bar{g}, \qquad \bar{b}.$$

The names are arbitrary. They do not refer to visible color, and the three colored versions of a given quark have the same mass and the same obvious properties. But they are distinct states, and that distinction is what rescues Fermi statistics.

Here the lecture makes a historical analogy that is worth preserving. Spin itself had once played this role in atomic physics: when two electrons seemed to occupy the same orbital state, the new quantum number of spin resolved the Pauli tension. Color plays the same conceptual role a second time. The strategy is the same: attach another label so that the particles are not really identical in all degrees of freedom. In the Δ^{++}, one red, one green, and one blue quark are enough to remove the contradiction.

The lecture then pivots from the existence of color to the dynamics built from it. Before doing so, it reminds us how economical QED is. The whole theory can be summarized by one basic vertex: an electron emits or absorbs a photon. For bookkeeping purposes one may even pretend that a photon behaves in some respects like an electron–positron pair. Not because it is literally made of one, but because it shares the neutral charge bookkeeping and the spin-one structure. That fiction is not especially useful in QED, but it becomes useful in QCD.

A quark is therefore organized as a three-component color state,

$$q = \begin{pmatrix} r \\ g \\ b \end{pmatrix}, \qquad \bar{q} = \begin{pmatrix} \bar{r} & \bar{g} & \bar{b} \end{pmatrix}. \tag{2.21}$$

A gluon is then introduced, at the lecture's level, as a color–anticolor bookkeeping object:

$$q\,\bar{q} \sim \begin{pmatrix} r\bar{r} & r\bar{g} & r\bar{b} \\ g\bar{r} & g\bar{g} & g\bar{b} \\ b\bar{r} & b\bar{g} & b\bar{b} \end{pmatrix}. \tag{2.22}$$

This suggests nine gluons at first sight. The lecture then immediately adds the caveat that one diagonal superposition is not an independent state. The full group-theoretic derivation is postponed, but the minimal cleaned statement is

$$3 \otimes \bar{3} = 8 \oplus 1. \tag{2.23}$$

The lecture's phrasing is more heuristic: it looks like nine, but really one particular diagonal sum is "a nothing." That is enough for the present level.

The primitive quark–gluon interaction is the color-changing emission process

$$q^a \longrightarrow q^b + g^{a\bar{b}}. \tag{2.24}$$

For example,

$$r \longrightarrow g + (r\bar{g}). \tag{2.25}$$

This is the QCD analogue of photon emission in QED, but now the emitted quantum itself carries color bookkeeping labels. That is why the lecture keeps emphasizing the rule: follow the lines. Track the color through the diagram. If a line turns around in time, reinterpret it as an antiparticle. At this level that is essentially the whole computational grammar.

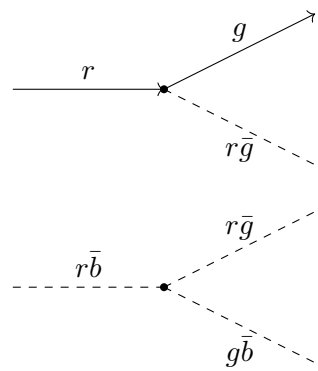


Figure 2.4: Schematic QCD vertices in the spirit of the lecture. Above: a quark changes color and emits a gluon. Below: a gluon turns into two gluons. The labels are bookkeeping labels, and the operative rule is to follow the color lines.

The upper diagram is the direct analogue of the QED vertex. The lower diagram is the new ingredient. Because photons do not carry electric charge, they do not interact with one another at the primitive level. Gluons do. The lecture constructs this by treating the gluon as a quark–antiquark-like color flow and allowing that flow itself to split. The result is a three-gluon vertex. Once that exists, gluons can be exchanged not only between quarks but also between gluons.

That is the decisive novelty of QCD. Quarks bind by exchanging gluons, but gluons themselves can also bind through gluon exchange. The lecture emphasizes three consequences. First, QCD can contain objects with no quarks at all, only gluonic degrees of freedom. Second, the dynamics of the gluon field is non-linear: one gluon wave does not simply pass through another without effect, and even different parts of a single gluon field configuration can act back on one another. Third, the bookkeeping is more intricate than in electrodynamics, which is why the lecture does not try to canonize every color-labeled sketch on the board. The stable lesson is the line-following rule, not every transient chalk label.

The lecture closes with a heuristic coda on running parameters. Imagine a small mass m attached to a soft extended blob. Push very slowly, and the whole structure moves together; the effective inertia is the mass of the full composite. Shake very rapidly, and the extended part does not have time to respond; then one probes only the small core. The observed mass depends on the scale of the probe. This is the lecture's route to the statement that the effective parameters of a quark are scale dependent:

$$m_q = m_q(\mu), \qquad Q_q = Q_q(\mu). \tag{2.26}$$

This is not a renormalization-group derivation. It is a physical analogy, meant only to explain why the "mass of a quark" and even the effective charge one attributes to it depend on how hard one probes the hadron.

The lecture deliberately refuses to close the subject here. It tells us instead what comes next: confinement, hadron structure, and the deeper reason color is never seen in isolation. For the moment the lesson is more modest. QCD works by color, gluons are the carriers of that interaction, and unlike photons they interact with one another.

2.8 Summary

The chapter has followed the lecture's intended motion from analogy to contradiction to resolution. QCD was first introduced by comparison with QED, but the real conceptual preparation was the return to spin. From the commutators and the $2\ell + 1$ rule we were led to the two-spin puzzle, then to the triplet and singlet, then to the three-spin decomposition. Only after that did the lecture reinterpret the same mathematics as isospin.

With isospin in hand, the proton and neutron emerged as the isospin-$\frac{1}{2}$ doublet, while the Δ quartet supplied the isospin-$\frac{3}{2}$ sector and, more importantly, the Pauli puzzle that forces color into the theory. Once color is admitted, the lecture can finally pass to QCD proper: color-labeled quarks, color–anticolor gluon states, the basic quark–gluon vertex, the three-gluon interaction, and the non-linear dynamics that distinguishes QCD sharply from electrodynamics.

The chapter therefore ends exactly where the lecture wants to end: with a mathematically guided intuitive picture, not with a completed theory of hadrons. The next step is confinement. Here the purpose was to see why color had to be invented, what kind of force carrier the gluon is, and how the first narrative of QCD is assembled.